STACKIFICATION COMMUTES WITH INERTIA

THE STACKS PROJECT AUTHORS

Editorial note. Formula-composition proof: the body states both facts (stackification commutes with 2-fibre products; the inertia is a 2-fibre product by formula) and the conclusion is their forced composition; not a pointer to another version of the same statement.

Allowed standard prerequisites (editorial).

- Stackification and 2-fibre products. lemma-stackification-fibre-product-fibred-categories (same chapter, named). Provides the commuting equality for 2-fibre products used twice.
- Inertia as a 2-fibre product. categories-lemma-inertia-fibred-category (cross-chapter, named). Gives the formula expressing the inertia fibred category as a 2-fibre product over X x X.

Source reference. Original reference 034H: focused-seeds/geometry/S9-stk-stackification-inertia/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04Y1: sources/stacks-stacks/pinned-original-source.tex, source lines 2005–2005. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{X}$ be a fibred category over $\mathcal{C}$. The stackification of the inertia fibred category $\mathcal{I}_\mathcal{X}$ is inertia of the stackification of $\mathcal{X}$.*

Proof. This follows from the fact that stackification is compatible with 2-fibre products by Lemma 04Y1 and the fact that there is a formula for the inertia in terms of 2-fibre products of categories over $\mathcal{C}$, see Categories, Lemma 034H. □

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